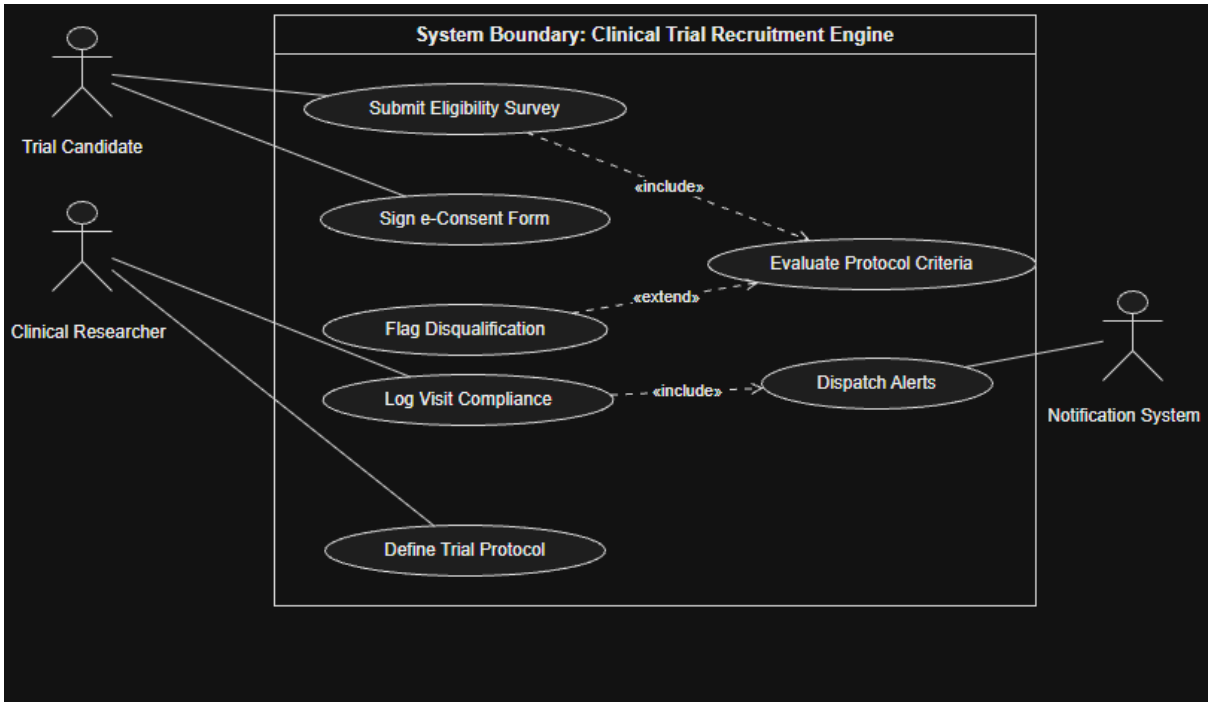
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall evaluate candidate survey responses against trial protocol inclusion/exclusion criteria to classify eligibility status.	High	**Pass:** Eligible candidate is advanced to consent, ineligible candidate is flagged within 2s.	Automates initial screening to save clinical staff time.
FR-002	Functional	The system shall capture, timestamp, and store electronic Informed Consent Forms (e-ICF) with digital signatures.	High	**Pass:** Unsigned form blocks progression, signed form generates a cryptographic audit record.	Ensures legal compliance before trial participation.
FR-003	Functional	The system shall allow the Clinical Researcher to log subject visit compliance and attendance status.	Medium	**Pass:** Researcher logs visit, missed visit flags the candidate's profile and triggers an alert.	Prevents protocol deviations and missing data points.
FR-004	Functional	The system shall dispatch automated alerts and notifications for appointment reminders and status updates.	High	**Pass:** Notification dispatched via SMS/Email within 60	Enhances subject retention and minimizes manual follow-ups.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
				seconds of a trigger event.	
FR-005	Functional	The system shall allow Clinical Researchers to define custom trial protocols (criteria rules and visit schedules).	High	**Pass:** Validated protocol configuration saves to the active trial roster.	Provides flexibility to support diverse therapeutic studies.
NFR-001	Nonfunc.	The recruitment database must comply with FDA 21 CFR Part 11 electronic record standards including cryptographic audit trails.	High	**Pass:** All edits show user, timestamp, and prior value, zero unauthorized record alterations.	Mandatory regulatory compliance for clinical research.
NFR-002	Nonfunc.	The system shall complete the eligibility criteria evaluation for any submitted survey in under 3 seconds.	High	**Pass:** A timing log shows < 3 seconds from survey submission to status generation.	Ensures quick processing and a smooth candidate experience.



Use Case: Submit Eligibility Survey Primary Actor: Trial Candidate

Brief Description: Allows an authenticated candidate to apply for a trial and complete an eligibility screening survey, which the system automatically evaluates against the defined protocol.

Preconditions:

1. The Clinical Researcher has executed the Define Trial Protocol use case and the trial is active.
2. The Trial Candidate is authenticated into the candidate portal.

Postconditions:

- Success Guarantee: Candidate status is updated to “Eligible”, candidate is cleared to proceed to the Sign e-Consent Form phase.
- Minimal Guarantee: Candidate is disqualified gracefully with a logged reason, no consent state changes occur.

Main Success Scenario:

1. Trial Candidate selects the “Apply for Trial” option from the portal.
2. System renders the Submit Eligibility Survey form.
3. Candidate inputs demographic and medical history responses and submits the form.

4. System executes the “include” Evaluate Protocol Criteria use case to validate the survey responses against the predefined trial rules.
5. System determines full eligibility and updates the candidate's status to “Eligible - Pending Consent”.
6. System automatically redirects the candidate to the Sign e-Consent Form use case.
7. Use case ends successfully.

Alternate Flow: 4a. Candidate Fails Screening Criteria

- 4a1. System detects one or more disqualifying conditions (e.g., conflicting medication or age out of bounds) during the evaluation.
- 4a2. System triggers the “extend” Flag Disqualification use case.
- 4a3. System updates the candidate's record to “Ineligible” and logs the specific failed criteria to the researcher's backend view.
- 4a4. System displays a polite disqualification message to the candidate and provides contact info for alternative trials.
- 4a5. The application process is safely aborted, and the use case ends.